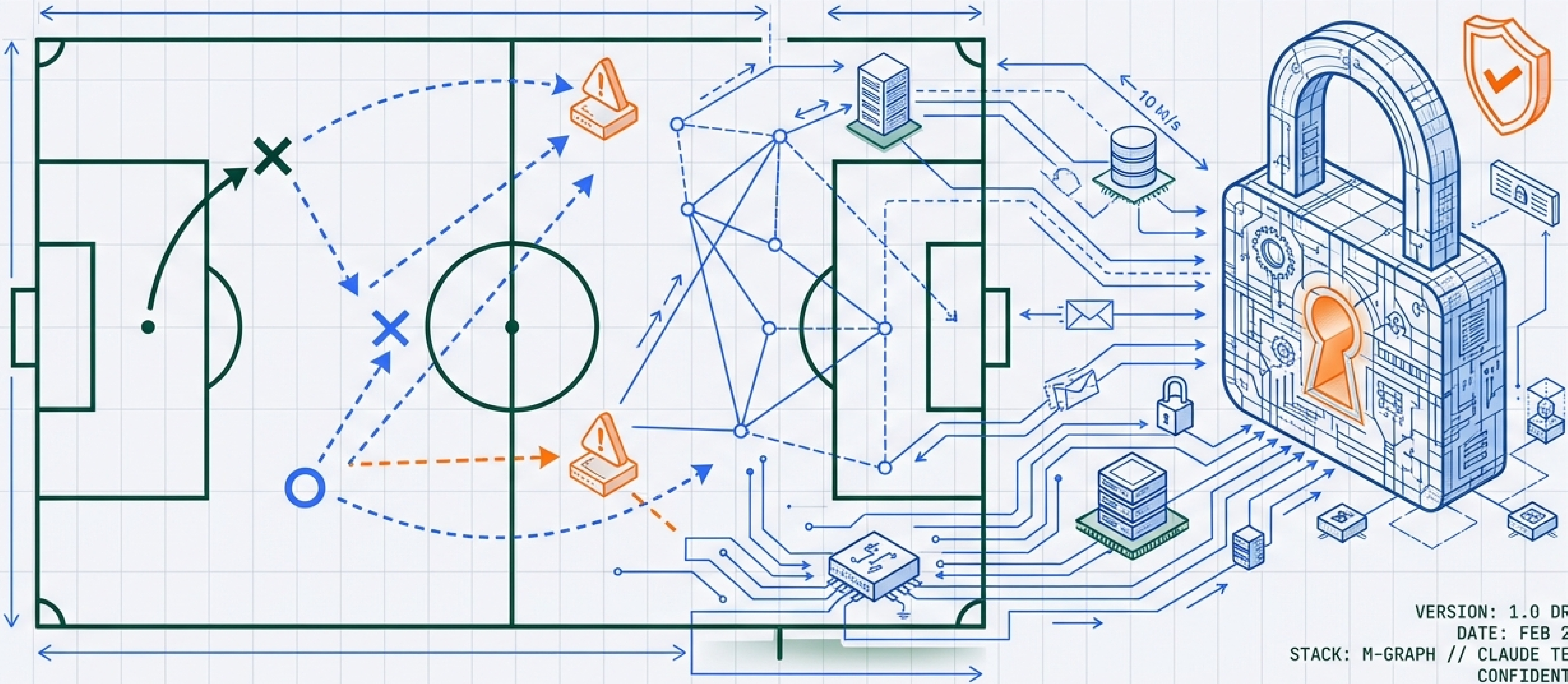
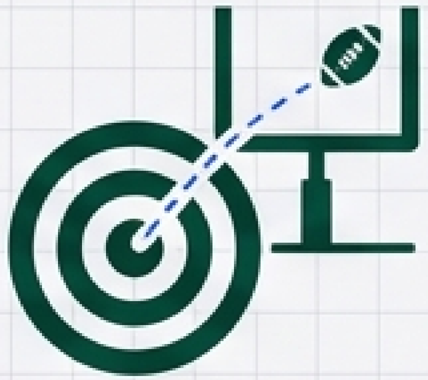
FOOTBALL PILATES SCHEDULER

A Privacy-First Coordination Engine for Local Sports Teams



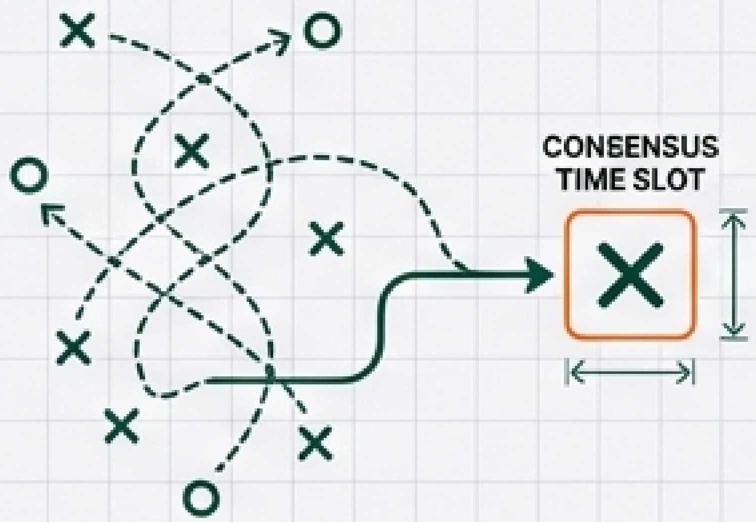
VERSION: 1.0 DRAFT
DATE: FEB 2026
STACK: M-GRAPH // CLAUDE TEAMS
CONFIDENTIAL

EXECUTIVE SUMMARY



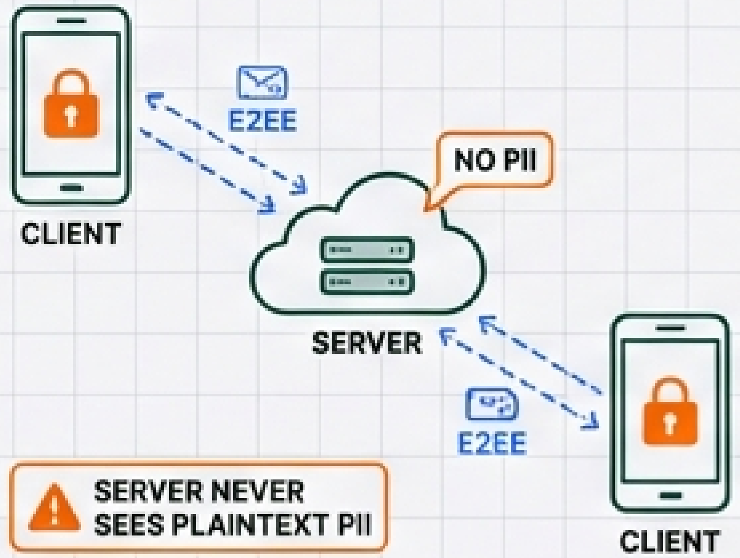
THE MISSION

Coordinate Pilates classes for a network of football friends in Chiswick. The tool resolves the chaos of multiple WhatsApp groups to find a consensus time slot.



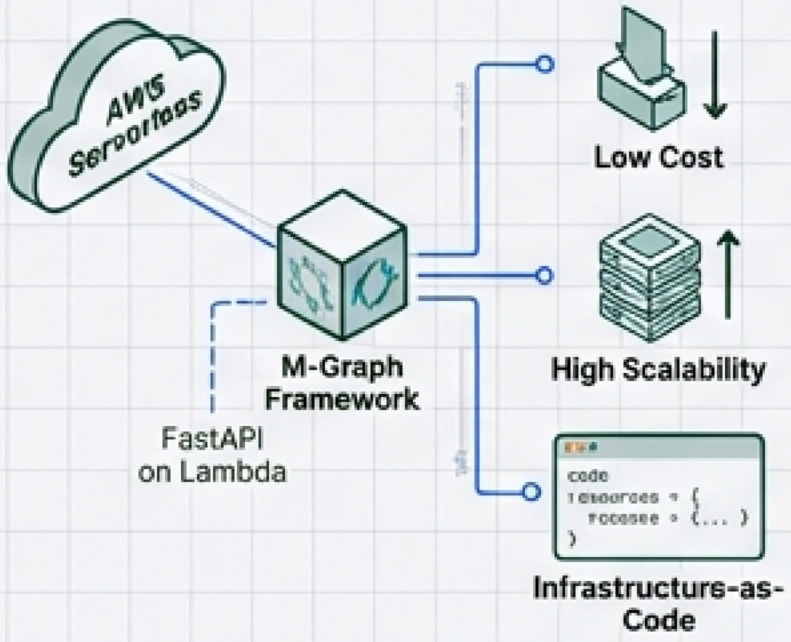
THE USP

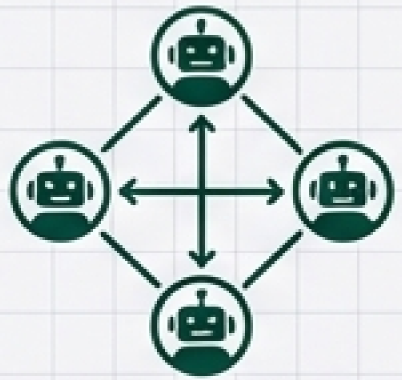
A 'Doodle'-like utility built with Signal-grade privacy. Features client-side End-to-End Encryption where the server never sees plaintext PII.



THE STACK

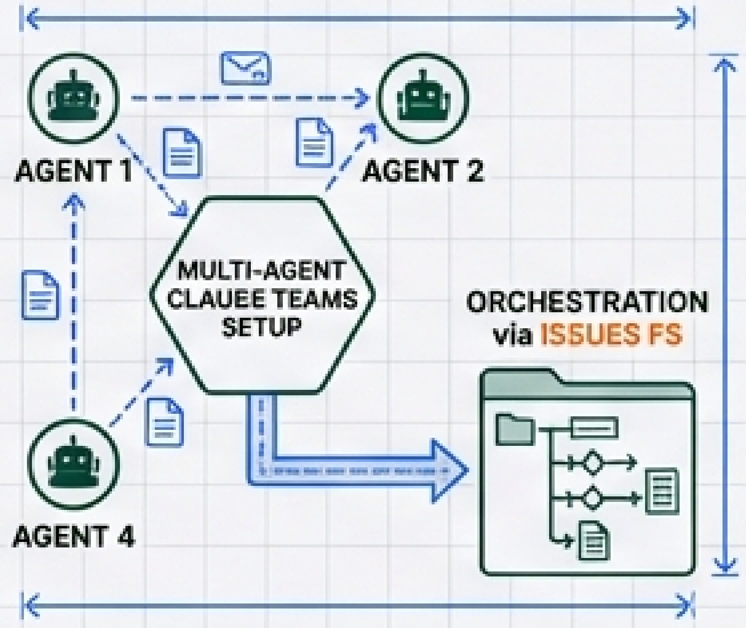
Built on AWS Serverless infrastructure using the M-Graph framework (FastAPI on Lambda). Ensures low cost, high scalability, and infrastructure-as-code.





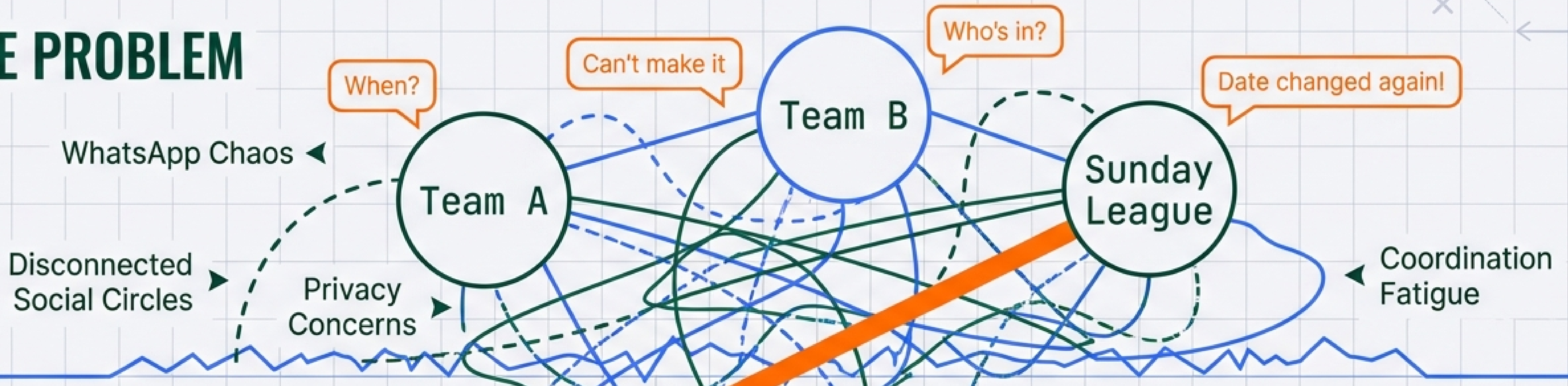
THE TEAM

Development executed entirely by an autonomous Multi-Agent Claude Teams setup, orchestrated via Issues FS.



CONTEXT: WHY BUILD THIS?

THE PROBLEM



THE SOLUTION

- **THE OPPORTUNITY:** A Chiswick studio offers a dedicated 'footballers' Pilates' class to improve core strength and injury management.
- **THE USER:** The Organiser (Age 35–60). Bridges multiple groups. Needs a lightweight tool.
- **THE FIX:** A bespoke web app to identify the optimal time slot without the admin overhead.

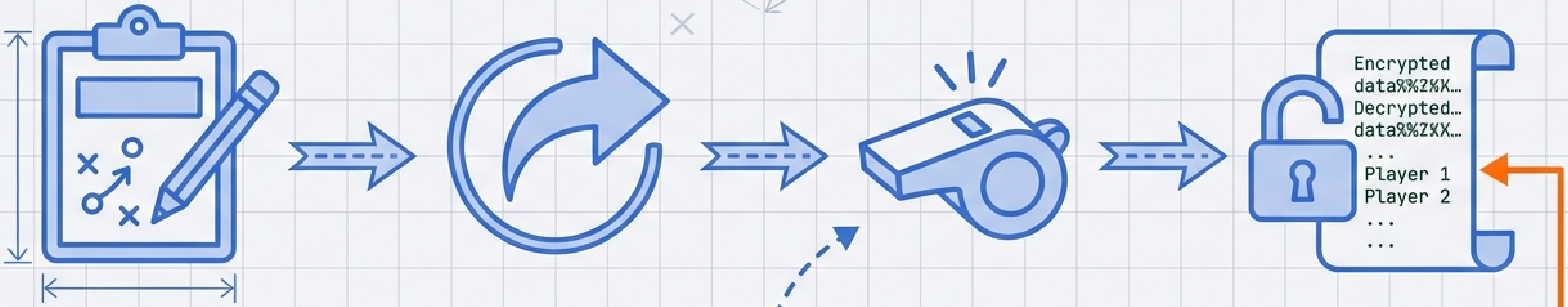
THE ORGANISER EXPERIENCE (ADMIN)

CREATE POLL

SHARE LINK

LIFECYCLE CONTROL

CLIENT-SIDE DECRYPTION



Create Poll: Define dates/slots. Browser generates Public/Private keypair.

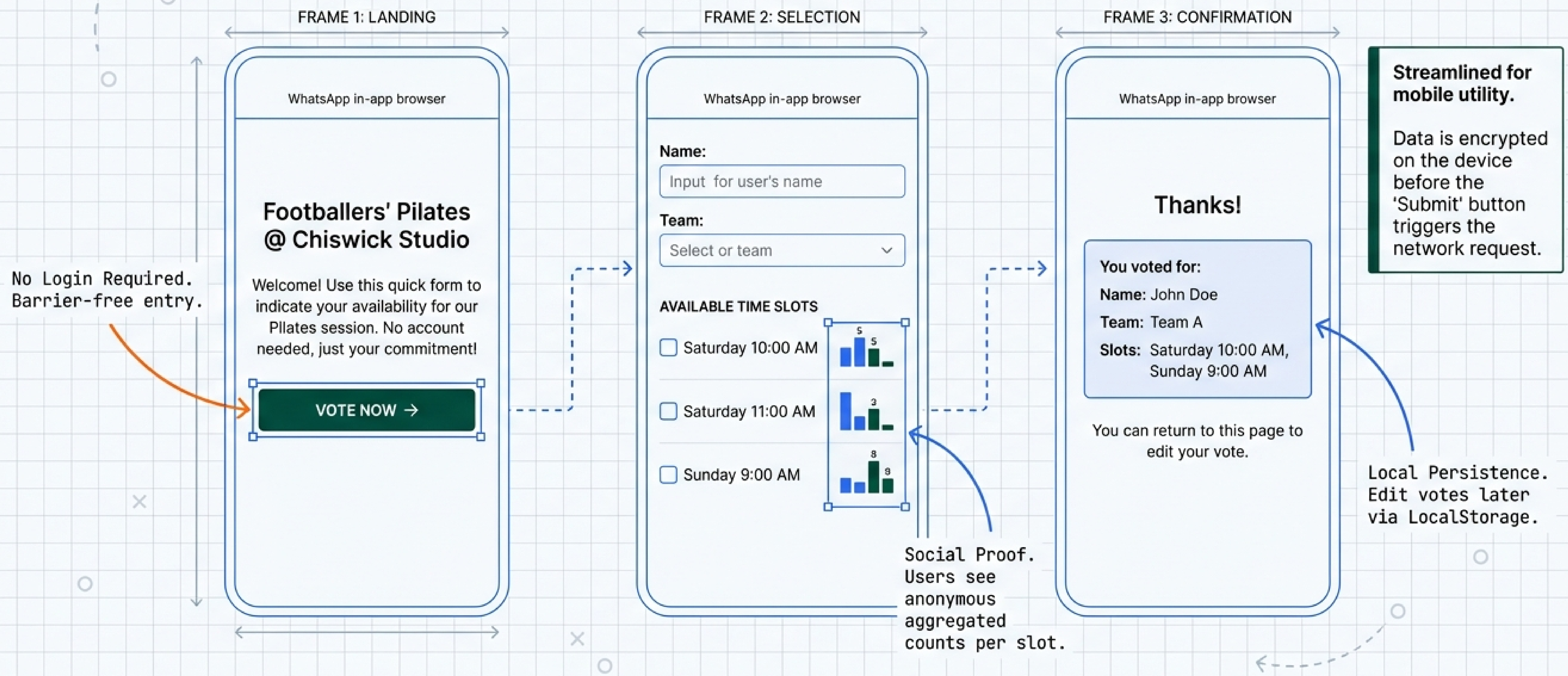
Share Link: Distribute URL via WhatsApp. Public key embedded in link.

Lifecycle Control: Start/Stop voting periods. Monitor total vote count.

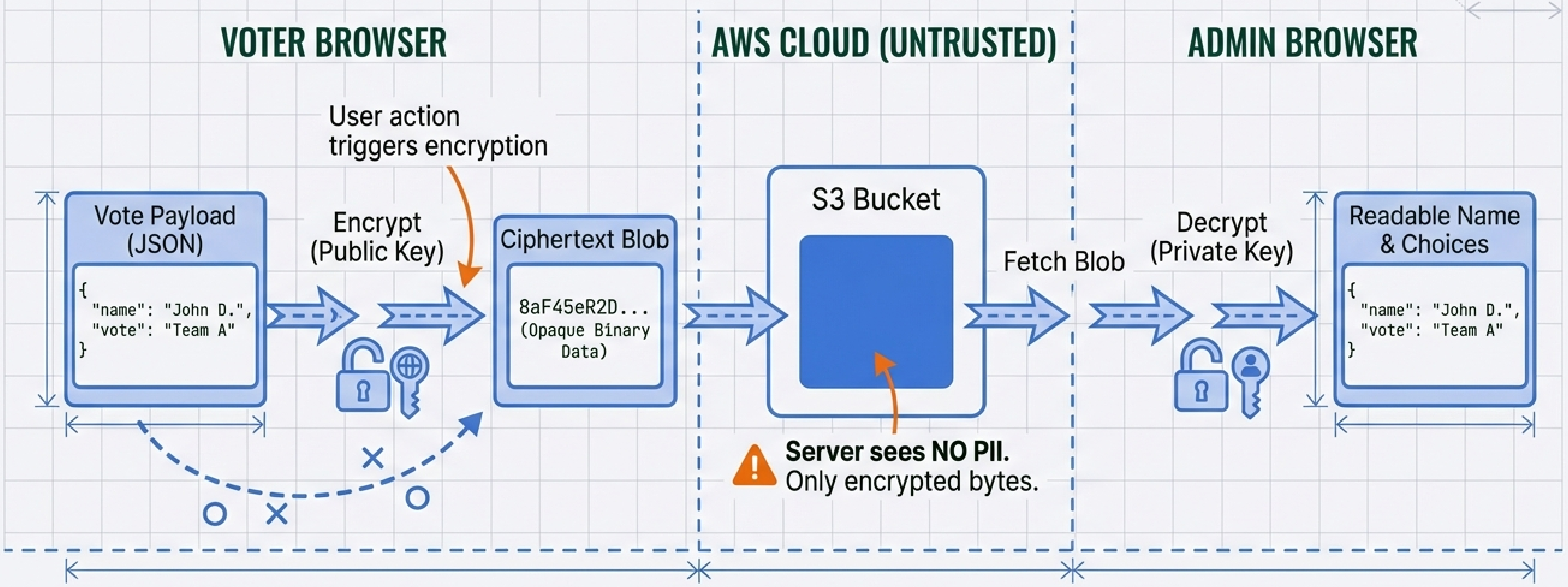
Client-Side Decryption: Dashboard fetches encrypted blobs. Decrypts locally with Private Key.

KEY INSIGHT: Only the holder of the private key can read the names on the list. The server remains blind.

THE VOTER EXPERIENCE



SECURITY ARCHITECTURE: ZERO-KNOWLEDGE



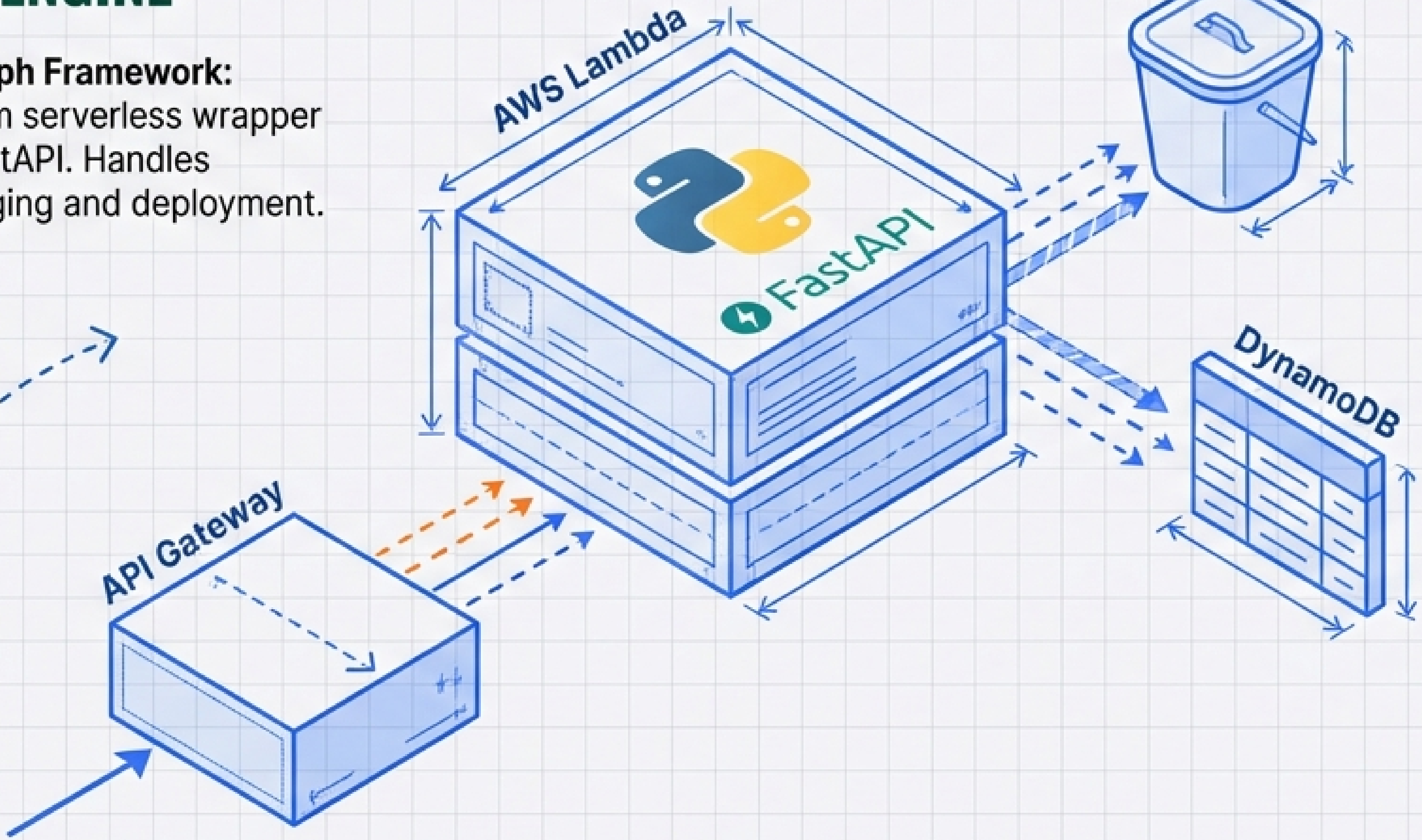
STANDARDS: Evaluating RSA-OAEP vs. X25519 (ECDH) + AES-GCM via Web Crypto API.

CORE PRINCIPLE: What the server can't read, the server can't leak.

BACKEND INFRASTRUCTURE: M-GRAPH

THE ENGINE

M-Graph Framework: Custom serverless wrapper for FastAPI. Handles packaging and deployment.

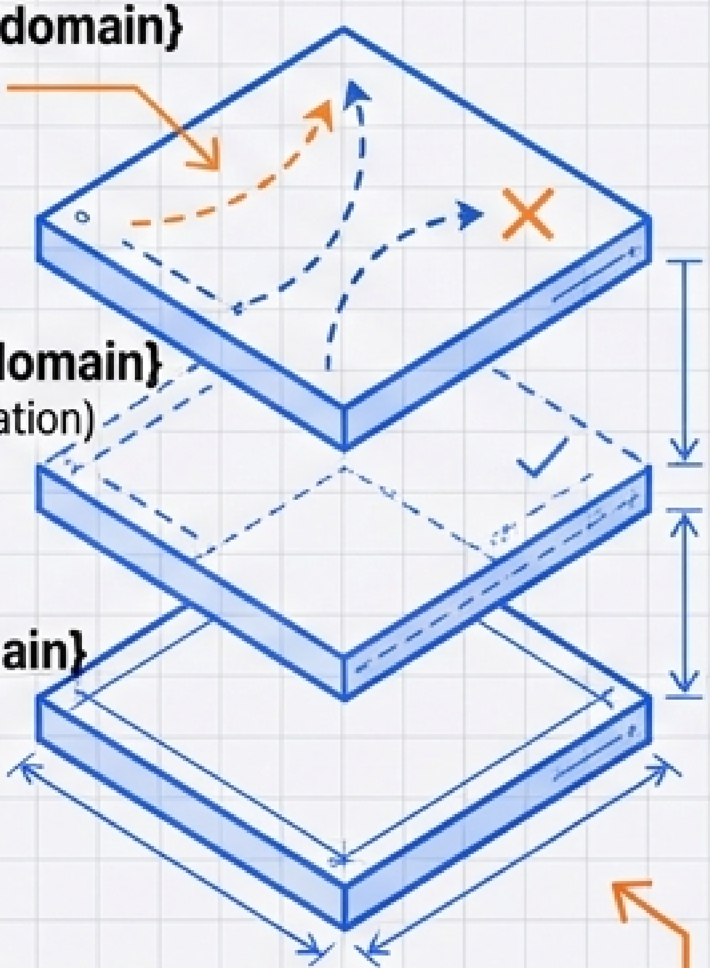


DEPLOYMENT STRATEGY

1. dev-pilates.{domain}
(Unstable/Active)

2. qa-pilates.{domain}
(Integration/Validation)

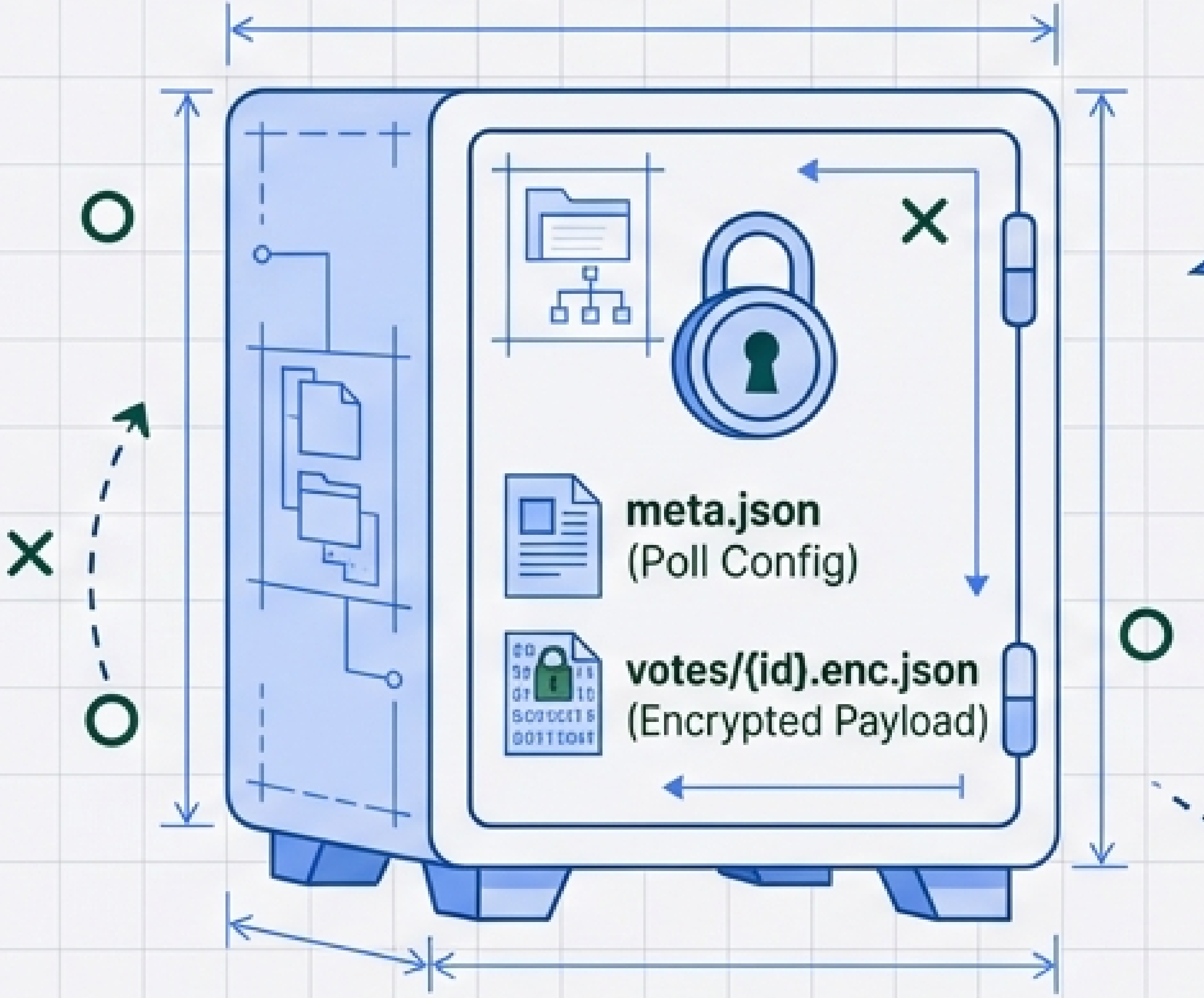
3. pilates.{domain}
(Live Production)



Infrastructure-as-Code ensures rapid and consistent environment promotion. Inter Regular

DATA STRATEGY: HYBRID STORAGE MODEL

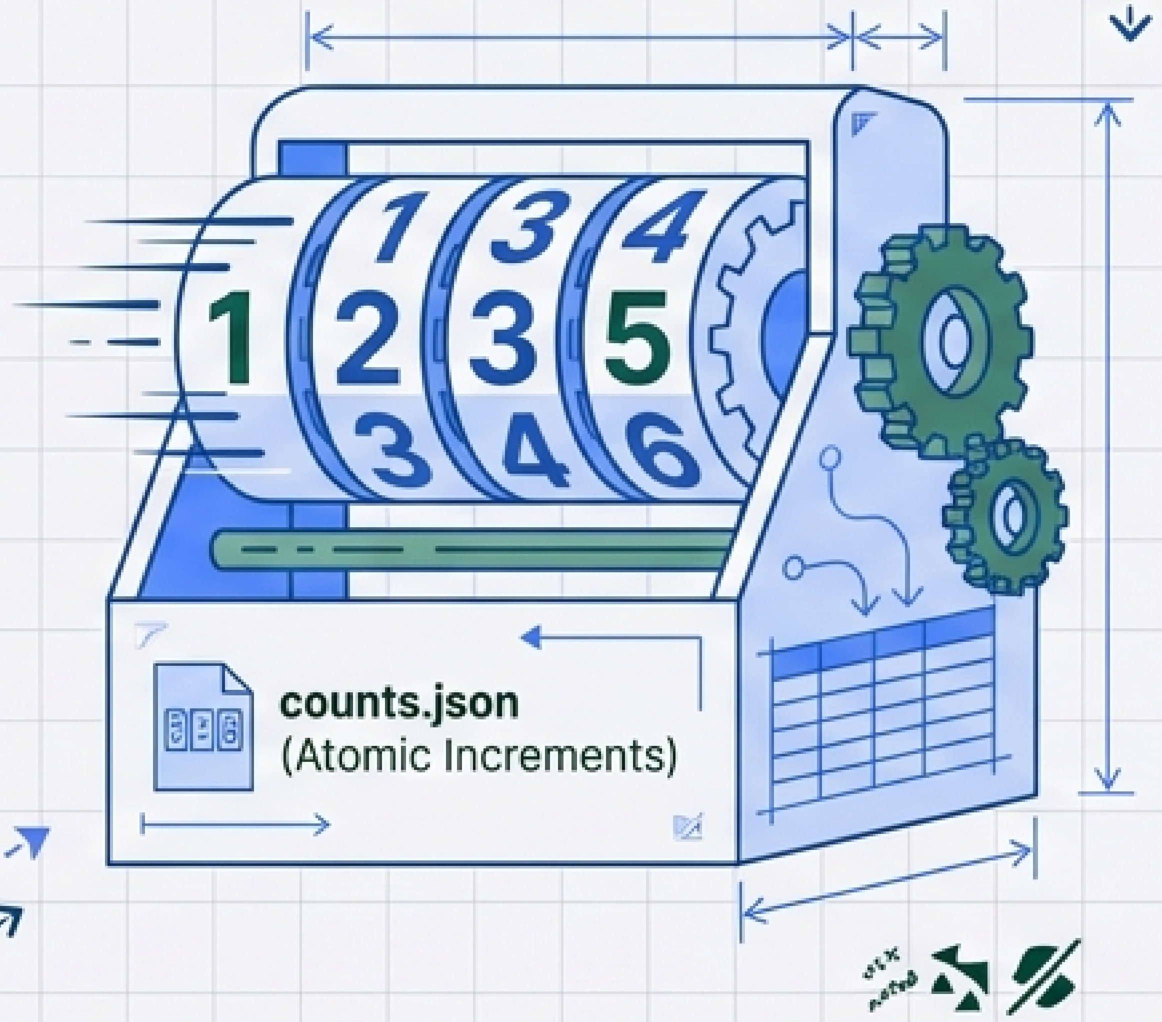
AMAZON S3



The Database of Record.
Encryption at Rest (SSE-S3/KMS).

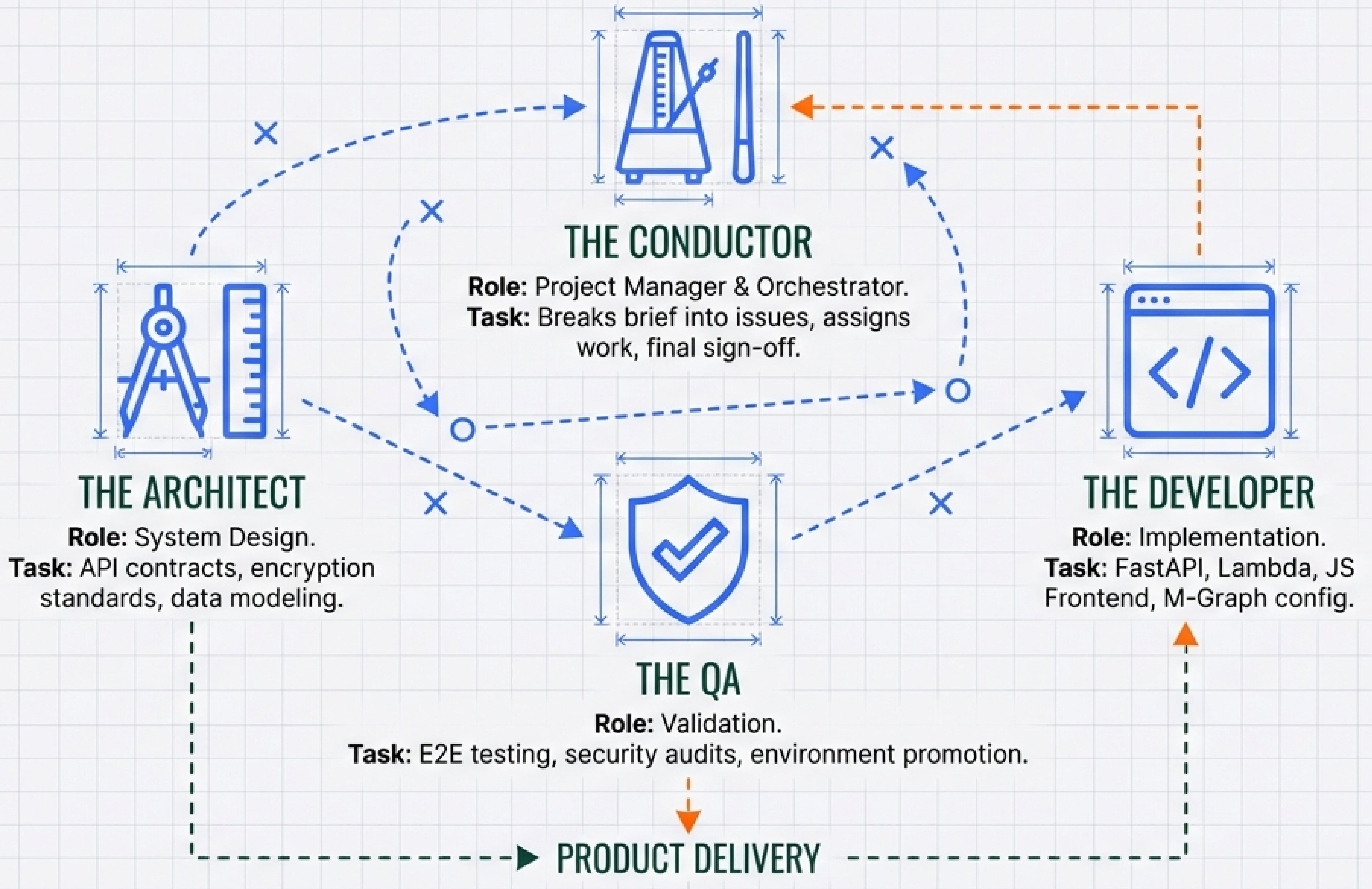
Hybrid Approach balances the Organiser's preference for file-based storage with the need for concurrent write safety.

AMAZON DYNAMODB

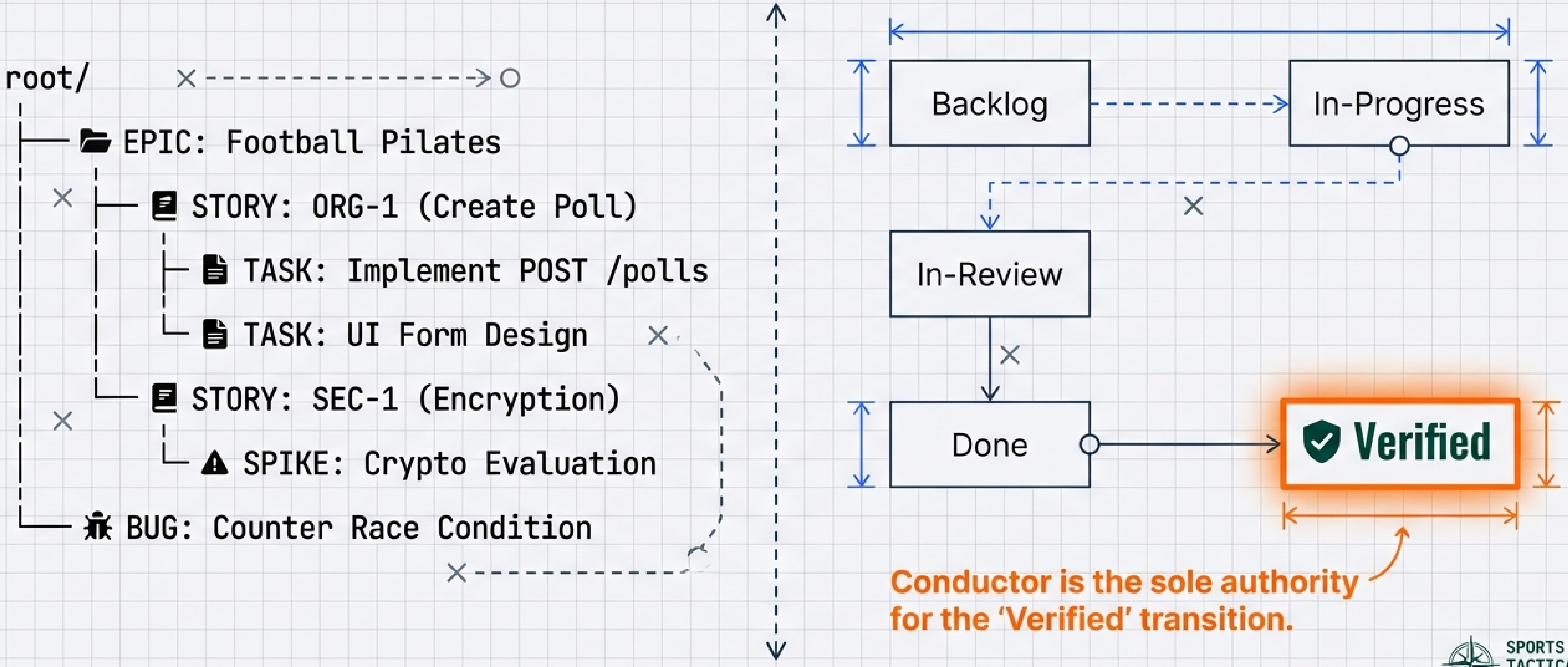


The Atomic Counter. Prevents race conditions on popular slots.

THE BUILDERS: CLAUDE TEAMS MULTI-AGENT SETUP

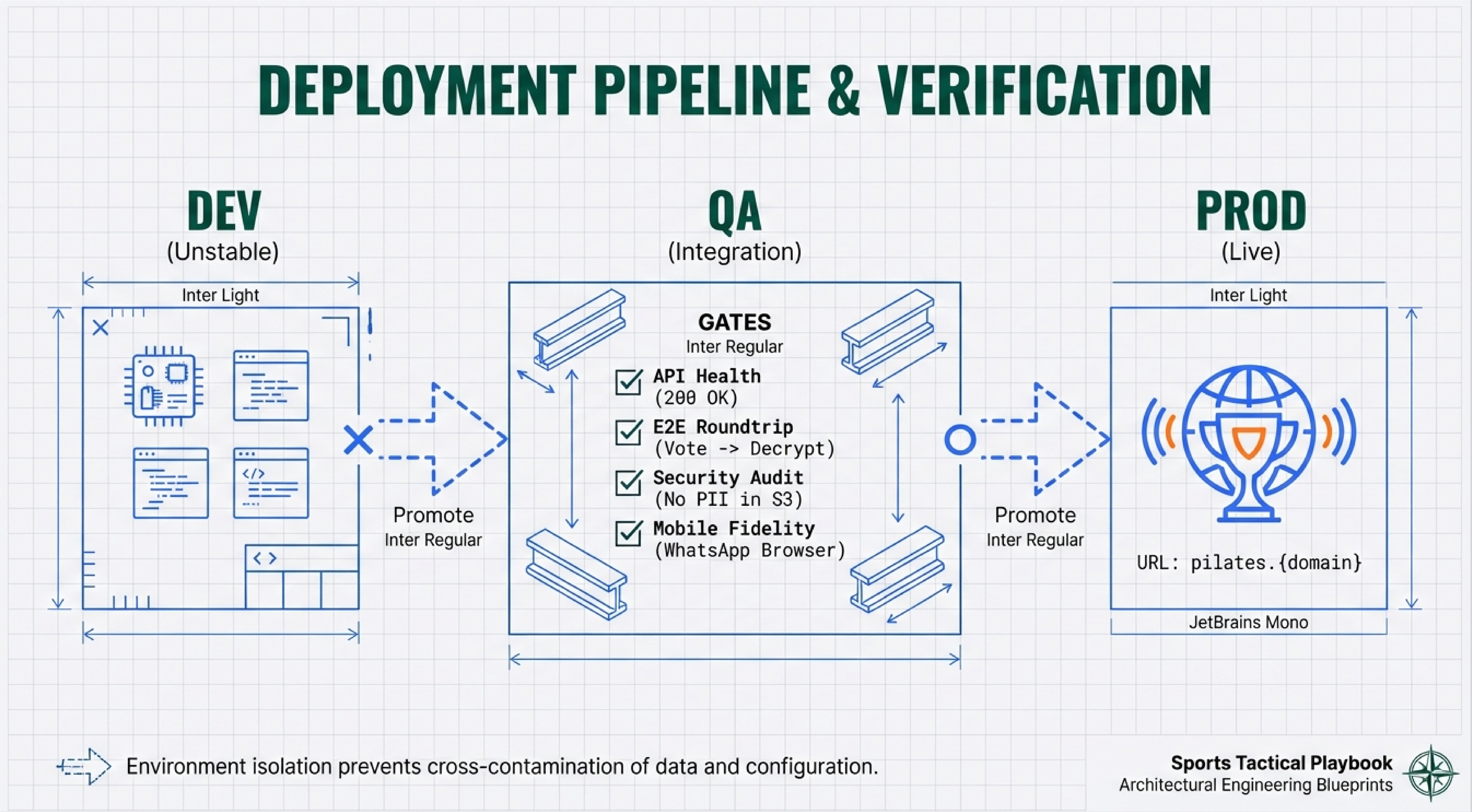


ORCHESTRATION & WORKFLOW: ISSUES FS



SPORTS TACTICAL PLAYBOOK

Architectural Engineering Blueprints


DEPLOYMENT PIPELINE & VERIFICATION
DEV
(Unstable)
Inter Light
Promote
Inter Regular
QA
(Integration)
GATES
Inter Regular
API Health
(200 OK)
E2E Roundtrip
(Vote -> Decrypt)
Security Audit
(No PII in S3)
Mobile Fidelity
(WhatsApp Browser)
Promote
Inter Regular
PROD
(Live)
Inter Light
URL: pilates.{domain}
JetBrains Mono
Environment isolation prevents cross-contamination of data and configuration.
Sports Tactical Playbook
Architectural Engineering Blueprints

KEY TECHNICAL DECISIONS (SPIKES)

CRYPTO SCHEME

Option A: RSA-OAEP
(Native, Size Limits)

Option B: X25519 + AES-GCM
(Flexible, Complex)

Architect to Execute Spike.

COUNTER ATOMICITY

Challenge: Race conditions during simultaneous votes.

Trade-off: S3 Conditional Puts vs. DynamoDB Atomic Increments.

Reliability Analysis.

FRONTEND TECH

Option A: Vanilla JS
(No deps)

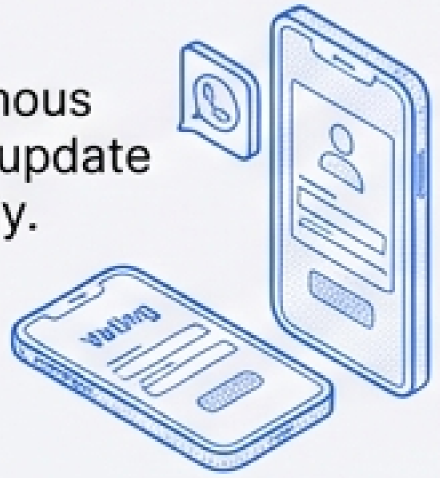
Option B: Preact
(Lightweight framework)

Evaluate bundle size vs. dev speed.

SUCCESS CRITERIA & ACCEPTANCE

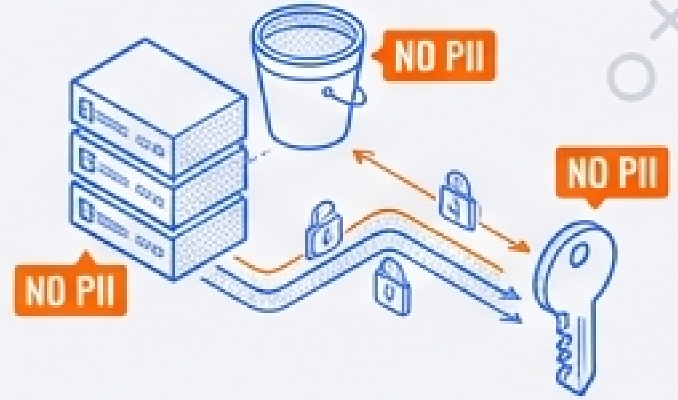
FUNCTIONAL

- ☑ Organiser coordinates time via WhatsApp link.
- ☑ Voters participate without login.
- ☑ Anonymous counts update correctly.



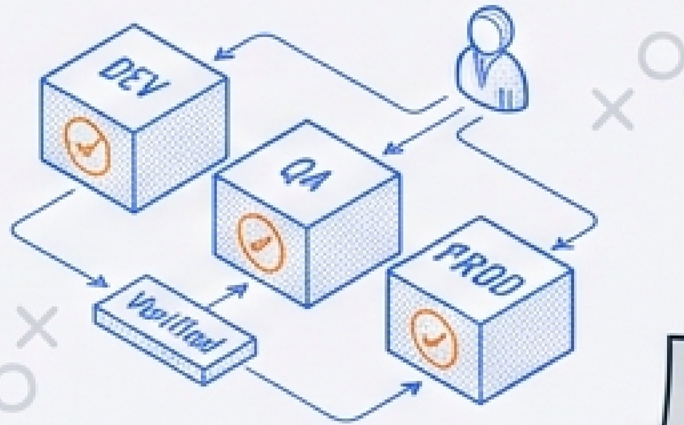
SECURITY

- ☑ Server/S3 NEVER holds PII.
- ☑ Only Private Key holder views names.

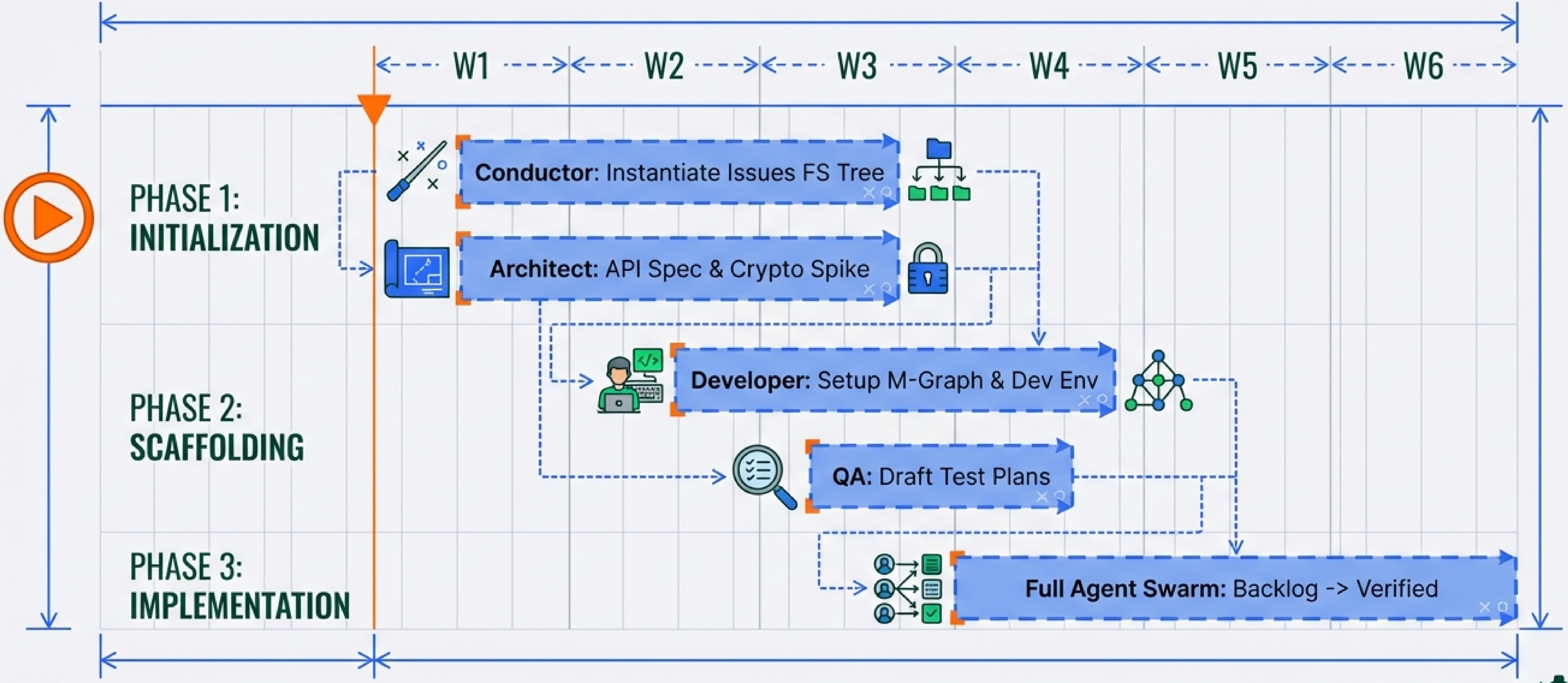


OPERATIONAL

- ☑ Dev, QA, Prod environments function independently.
- ☑ All Issues FS items marked 'Verified' by Conductor.



ROADMAP & NEXT STEPS



Sports Tactical Playbook
Architectural Engineering Blueprints

FOOTBALL PILATES SCHEDULER

Bringing the squad together, securely.